

Monte Carlo integration

Introduction

In statistical mechanics, observables are expressed as ensemble averages, which require the evaluation of high-dimensional integrals (often 10^6 dimensions or more) [1]. Ensemble averages are used to predict thermodynamic properties of materials and liquids [2]. For realistic many-particle systems, these integrals generally cannot be found analytically [2]. Numerical methods evaluate the integrand on a regular grid, but these methods scale exponentially with dimension, making them impractical for high-dimensional integrals [3].

Monte Carlo (MC) integration is an alternative approach that uses stochastic sampling. Rather than evaluating the integrand on a regular grid, points are drawn randomly, and integrals are approximated by sample averages. Importance sampling draws samples from regions with a larger contribution to the integral at a greater frequency, reducing variance [4]. If sampling directly from a target distribution is not possible, Markov chain [5] methods such as Metropolis-Hastings [6] can generate samples from the target distribution.

We apply MC integration to a system of harmonic oscillators in the canonical ensemble [1]. The average potential energy is known by the equipartition theorem [1], providing a benchmark for comparison. We then evaluate the limitations and accuracy of direct importance sampling and the Metropolis-Hastings algorithm as a way of calculating ensemble averages.

Theory

Further derivations are provided in [7].

The Harmonic Oscillator

We apply MC techniques to a many-particle harmonic oscillator in the canonical ensemble [1].

The harmonic oscillator is the simplest model of small vibrations about an equilibrium. The canonical ensemble implies a fixed number of particles K , temperature T , and volume V .

Consider a force constant k , Boltzmann constant k_B , and configurations $q = (q_1, q_2, \dots, q_K)$. Each configuration occurs with the Boltzmann density, $p(q) = \frac{1}{Z} \exp(-\beta U(q))$ [2] with $\beta = (k_B T)^{-1}$, Z as the partition function $Z = \int_{\mathbb{R}^K} \exp(-\beta U(q)) dq$ and $U(q)$ as the potential energy,

$$U(q) = \sum_{i=1}^K \frac{1}{2} k q_i^2. \quad (1)$$

The observable of interest is the ensemble average potential energy given by [2],

$$\langle U \rangle = \int_{\mathbb{R}^K} U(q) p(q) dq. \quad (2)$$

This can be solved by application of the equipartition theorem [1], which yields,

$$\langle U \rangle = \frac{K}{2} k_B T. \quad (3)$$

This is a benchmark to compare our estimates with. Equipartition gives a solution because the harmonic oscillator has a quadratic potential, but for most realistic interacting many-particle systems this is not the case, such as particle systems under the Lennard-Jones potential [2], hence the results are not fully representative of the general case.

Basic Theory

We give an overview of MC integration built from [2] [4] [6] [7]. Consider an integral I ,

$$I = \int_{\Omega} f(x) dx, \quad (4)$$

where $x = (x_1, \dots, x_N)$ is an N -dimensional point in the domain $\Omega \in \mathbb{R}^N$, and $f(x)$ is the integrand over Ω . Let $p(x) \geq 0$ be a normalized probability density function (PDF). We then rewrite I as

$$I = \int_{\Omega} \frac{f(x)}{p(x)} p(x) dx = \mathbb{E}_p[h(X)], \quad (5)$$

where $h(x) = \frac{f(x)}{p(x)}$ and $X \sim p(x)$. By drawing τ samples $x^{(1)}, x^{(2)}, \dots, x^{(\tau)}$, an estimate $\hat{I}$ is given,

$$\hat{I} = \frac{1}{\tau} \sum_{j=1}^{\tau} h(x^{(j)}). \quad (6)$$

For independent samples, the error of $\hat{I}$ is given by $\sigma(h)/\sqrt{\tau}$, where $\sigma^2(h) = \text{Var}_p[h(X)]$ [4]. An estimator for the variance is given by $S^2(h)$,

$$S^2(h) = \frac{1}{\tau} \sum_{j=1}^{\tau} h(x^{(j)})^2 - \hat{I}^2. \quad (7)$$

The error of $\hat{I}$ decreases as $\tau^{-1/2}$, independent of dimension N , whereas deterministic numerical quadrature scales as M^N , where M is the number of grid points per dimension [3]. For a system of 100 particles with 2 degrees of freedom each the ensemble average requires a 200-dimensional integral [1], making grid-based methods impractical [3]. MC integration therefore avoids dimensionality growth and offers a useful method for finding ensemble averages.

Importance Sampling

The error $\sigma(h)/\sqrt{\tau}$ of the estimator $\hat{I}$ depends on $\sigma(h)$. Importance sampling chooses $p(x)$ in order to minimize $\sigma(h)$ [4] and hence the error of $\hat{I}$.

The standard deviation $\sigma(h)$ and the error are minimized when $h(x)$ is constant. This occurs when $p(x)$ is similar in shape to $|f(x)|$ [4].

Fig.1 illustrates this effect for three different PDFs $p(x)$. Variance is largest when $p(x)$ poorly approximates $|f(x)|$. The variance is reduced by using a uniform PDF, but is minimized by selecting a PDF that closely approximates $|f(x)|$.

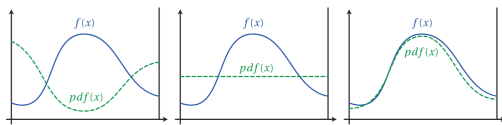


Figure 1: Three sampling distributions $p(x)$ relative to an integrand $f(x)$. The variance is minimized when $p(x)$ is close in shape to $f(x)$. Figure from [8].

Metropolis–Hastings algorithm

When evaluating ensemble averages [1], it's often impractical to sample directly from the target (Gibbs [1]) distribution, since normalization requires a high-dimensional integral for the partition function (Z) [2] (For 100 particles with 2 degrees of freedom each a 200-dimensional integral). Instead, the Metropolis-Hastings algorithm constructs a Markov chain [5] with an invariant distribution equal to the target distribution $p(x)$ [6]. A Markov chain is a sequence of random configurations $\{x^{(t)}\}$, where the next state $x^{(t+1)}$ depends only on the current state $x^{(t)}$ [5]. The algorithm generates a sequence of configurations that approach the target distribution, allowing ensemble averages to be estimated without evaluating the partition function.

Algorithm 1 Metropolis–Hastings algorithm

Start from $x^{(0)}$, iterate for $t = 1, 2, \dots, \tau$.

1. Sample a trial point $x' \sim T(x' | x^{(t-1)})$.
2. Compute the acceptance probability,

$$\alpha(x' | x^{(t-1)}) = \min(1, \frac{p(x')T(x^{(t-1)} | x')}{p(x^{(t-1)})T(x' | x^{(t-1)})}).$$

3. With probability $\alpha(x' | x^{(t-1)})$, set $x^{(t)} = x'$, otherwise set $x^{(t)} = x^{(t-1)}$.

In Alg.1, there is a choice for the starting point $x^{(0)}$ and the proposal distribution $T(x' | x)$ which gives the probability of sampling a trial point x' given the current point x . Here, successive points are correlated, hence samples are not independent.

Results

The simulation code is provided in [7].

Importance Sampling Simulations

To investigate importance sampling, the average potential energy (Eq.2) of a $K = 20, 100$ and 1000 particle harmonic oscillator with parameters $k = 10, T = 1$, and $k_B = 1$ was estimated by directly sampling from the Boltzmann distribution [1].

For each K , we drew $\tau = 10^5$ configurations $q^{(j)}$ by sampling each coordinate $q_i^{(j)}$ from the normal distribution $\mathcal{N}(0, \frac{k_B T}{k})$ corresponding to the

Boltzmann density $p(q) \propto \exp(-\beta U(q))$. For each configuration $q^{(j)}$, the potential energy $U(q^{(j)})$ (Eq.1) was evaluated. The mean and sample variance S^2 were computed using Eq.6 and Eq.7 with $U(q^{(j)})$ in place of $h(x^{(j)})$, and the standard error was estimated as $S/\sqrt{\tau}$.

Fig.2 shows the estimates of the mean potential energy $\langle U \rangle$ for $K = 20$ (a), 100 (b), and 1000 (c) as a function of the number of samples τ . The green line denotes the running mean, the black bars the estimated standard error, and the red line an equipartition prediction from Eq.3. The final estimates were $\langle U \rangle = 9.99 \pm 0.01$, 50.0 ± 0.02 , and 500 ± 0.07 , agreeing with the predicted values 10, 50, and 500. From Eq.2 we see that K determines the dimension of the integral. The observed error increases with K , but it decreases with sample size as $\tau^{-1/2}$ for all cases, illustrating that $\langle U \rangle$ is approached at the same rate for all cases, independent of dimension/the number of particles, as predicted by theory.

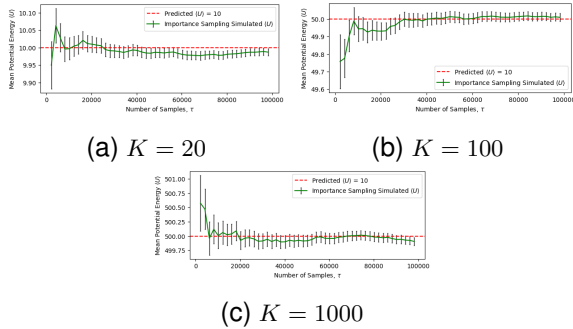


Figure 2: Average potential energy of 20 (a), 100 (b), and 1000 (c) harmonic oscillators estimated using direct importance sampling Monte Carlo integration as a function of the number of samples (solid, green), with standard error (bars, black) and the equipartition prediction (dashed, red).

Metropolis–Hastings Simulation

The Metropolis-Hastings algorithm(Alg.1) was used to approximate the average potential energy (Eq.2) of a $K = 20$ particle harmonic oscillator with parameters $k = 10$, $T = 1$, and $k_B = 1$.

For $\tau = 10^5$ steps, we started from the initial configuration $q^{(0)} = (0, 0, \dots, 0)$. At each step a trial configuration q' was given by $q'_i = q_i^{(t-1)} + \eta_i$, with $\eta_i \sim \mathcal{U}(-\delta/2, \delta/2)$, where $\mathcal{U}$ denotes the uniform distribution and δ is the pro-

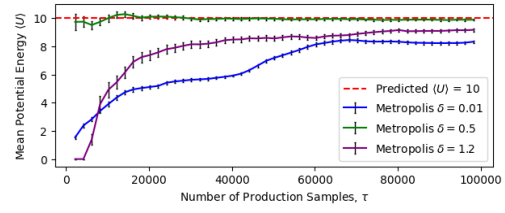


Figure 3: Average potential energy of harmonic oscillators estimated using the Metropolis-Hastings algorithm Monte Carlo integration as a function of the number of samples for different proposal step widths $\delta = 0.01$ (solid, blue), 0.5 (solid, green), 1.2 (solid, purple), with standard error (bars, black) and the equipartition prediction (dashed, red).

positional step. This defined a proposal distribution $T(q' | q^{(t-1)})$. With acceptance probability $\alpha = \min(1, \exp(-\beta(U(q') - U(q))))$. The potential energy $U(q^{(i)})$ was recorded at each step to produce samples, whose mean estimated $\langle U \rangle$. Since the samples are correlated, we estimated the uncertainty using block averaging [9]. The samples were divided into blocks of size 10, the mean was calculated within each block, and the variance and standard error were estimated from the variance of block means.

Fig.3 illustrates the estimated potential energy $\langle U \rangle$ for $\delta = 0.01, 0.5$ and 1.2 with blue, green and purple lines respectively. The standard error is shown with black error bars and the equipartition theorem prediction of 10(Eq.3) with a dashed red line. The plot for $\delta = 0.5$ best agrees with our benchmark with $\langle U \rangle = 10.0$ and an acceptance rate R (proportion of accepted steps) of 0.311, followed by $\delta = 1.2$ with $\langle U \rangle = 8.84$ and $R = 0.016$, then $\delta = 0.01$ is in worst agreement with $\langle U \rangle = 8.46$ and $R = 0.985$. This shows that the proposal distribution controls convergence. Large δ means moves are rejected, causing little exploration of the configuration space [1], while small δ means most moves are accepted, but with slow movement through the configuration space.

Conclusion

We showed both importance sampling and the Metropolis–Hastings algorithm to be effective methods for estimating ensemble averages in statistical mechanics, accurately reproducing the equipartition prediction for the harmonic oscilla-

tor. The results also confirmed the proposed ($\tau^{-1/2}$) convergence rate with respect to the number of samples τ , the main advantage of MC methods for high-dimensional integrals.

However, the harmonic oscillator was convenient for importance sampling as we sampled directly from its Boltzmann distribution. The efficiency of this technique is thus, not representative of other realistic many-body systems, such as those governed by a Lennard-Jones potential [2]. The Metropolis-Hastings algorithm's convergence was found to depend on its proposal distribution illustrating that techniques such as Langevin algorithms [6] should be employed to construct an optimal proposal distribution.

In the general context, MC integration allows for the realistic evaluation of material and thermodynamic properties of liquids, solids and gases by estimating ensemble averages [2]. Further work may consider systems whose ensemble averages aren't known analytically, for example systems governed by the Lennard-Jones potential [2]. Consideration of the optimal proposal distribution using extensions such as Langevin algorithms [6] may also provide improved results.

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